A LANGUAGE. AN INSTRUMENT. AN EXPERIMENT.

Kazakh music. *A new way to listen.*

Can adapting a pretrained model with Kazakh singing data produce music we prefer?

Yeldar & Aisha

Music generation · Assignment 2

01 / THE CHALLENGE

Recognizable words. *Music worth hearing.*

01

Kazakh language

Can a model connect sung lyrics to the audio it generates?

02

Musical character

Melody, phrasing and accompaniment should work together.

03

Actual preference

The final test is what a listener chooses to hear again.

Our task is **music generation**, using sung text and audio. We are not training a speech recognizer.

02 / DATASET ENRICHMENT

More Kazakh singing. *A careful split.*

3,013

source segments
195 songs · 36 artists



2,316	334	327
Training	Validation	Test

2,977 retained
36 segments excluded during filtering and deduplication.

Split by artist + song identity
Artists can still occur across splits. This is not an unseen-artist test.

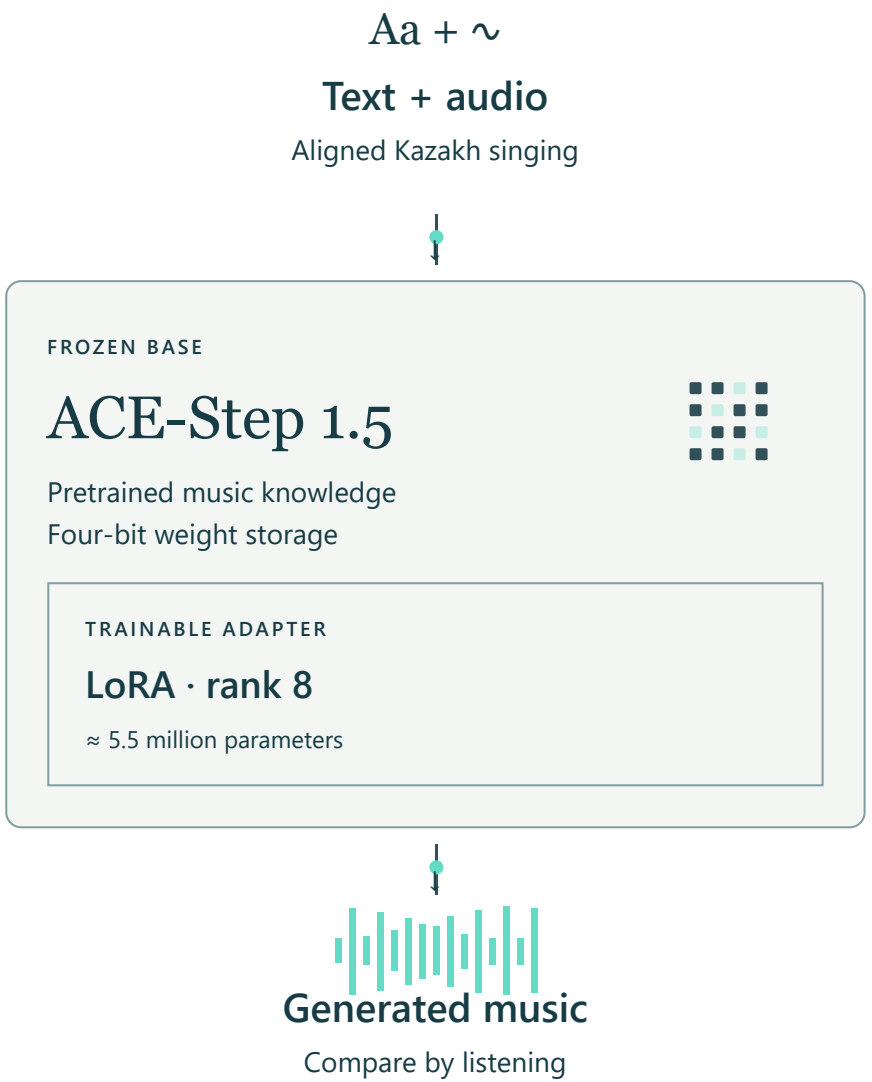
Aligned lyrics + isolated vocals
Useful for singing adaptation; limited evidence about full arrangements.

Source: Kazakh Songs ASR · gated, non-commercial academic use. Original recordings remain outside this presentation.

03 / WHAT WE TRAINED

Keep the foundation

Keep the foundation.
Learn a small adjustment.



We adapted an existing model. We did **not** train a foundation model from scratch.

04 / A COMPLETED LOCAL RUN

A full pass.
On a 4 GB laptop.

3,088
training updates

66.3 min
training time

1.43 GiB
peak allocated GPU memory



2,316 new clips, each used once
+ 772 updates over nine original music crops

0 held-out clip IDs in the training log
RTX 3050 Ti · frozen base · small adapter

Timeline replays recorded milestones, not live training. Timing excludes preparation, model loading and generation.

The number improved. *The music didn't convince us.*

The pilot lowered held-out flow loss. The owner still described the music as terrible.

HISTORICAL PILOT · TEST FLOW LOSS



−14.9%

numerical error

16 test clips from eight songs.
This is not a musical-quality score.

So we compared saved checkpoints, then changed **adapter strength** and listened again.

06 / THE LISTENING TEST

Same words. *Different influence.*

Same prompt, seed and 30-second duration.
Average playback levels matched.
Prerecorded AI-generated audio.

REFERENCE

Pretrained

Before adaptation



30-second prerecorded demonstration · play in the website

SELECTED DEMO

25% strength

Gentler adaptation



30-second prerecorded demonstration · play in the website

Waveforms come from these files. Highlighting follows actual playback. One sample plays at a time.

07 / THE LISTENING RESULT

Less influence.
A preferred example.



OWNER'S FOLLOW-UP CHOICE

Sample D

Final checkpoint · 25% adapter strength

FIRST COMPARISON

Pretrained beat the three full-strength checkpoints on the simple-vocal case.

STRENGTH FOLLOW-UP

The owner selected 25% strength for melody and accompaniment.

25% = adapter influence during generation.
All 2,316 eligible training clips were still used.

A positive preference on **one example**. No claim of a general improvement or better pronunciation.

08 / WHAT WE LEARNED

Train. Compare. *Listen carefully.*

What we demonstrated

Full-data adapter training on a 4 GB laptop, reproducible comparisons, and one preferred lighter-adapter example.

What comes next

More listeners, prompts and seeds. Native-speaker assessment. More suitable licensed accompaniment data.

Yeldar & Aisha

Questions? 2–3 minutes

Kazakh Songs ASR ↗ ACE-Step 1.5 ↗ Our code & evidence ↗ Local evidence

Dataset-enrichment route · confirm sung-language generation fits your instructor's NLP track. Source links require internet; this presentation does not.